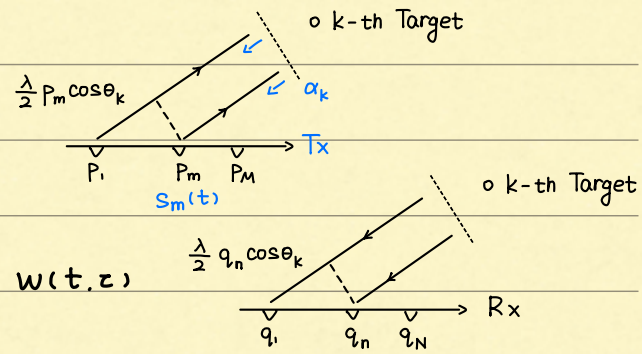
Fast Angle Estimator for MIMO Radar with Nonorthogonal Waveforms 2018

M 个发射，$s_m(t)$ $\qquad$ τ : slow time index / pulse index

阵列 $\mathbb{P} = \{p_1 \cdots p_M\}$

回波 $\underset{1\times1}{\alpha_k(t,\tau)} = r_k(\tau)\sum_{m=1}^{M} s_m(t)\, e^{-j\pi p_m \cos\theta_k}$

$$= r_k(\tau)\, a_t^T(\theta_k)\, s(t)$$



接收 $\underset{M\times1}{x(t,\tau)} = \sum_{k=1}^{K} \alpha_k \underset{M\times1}{a_r(\theta_k)} + w(t,\tau)$

$$= \sum_{k=1}^{K} r_k(\tau)\, a_r(\theta_k)\, a_t^T(\theta_k)\, s(t) + w(t,\tau)$$

观测

$$\underset{N\times1}{y_m(\tau)} = \int_{T_P} \underset{M\times1}{x(t,\tau)}\, \underset{1\times1}{s_m^*(t)}\, dt$$

不是共轭

$$= \sum_{k=1}^{K} r_k(\tau)\, a_r(\theta_k)\, \bar{a}_{t,m}(\theta_k) + n(t,\tau)$$

$$\bar{a}_{t,m}(\theta_k) = a_t^T(\theta_k) \int_{T_P} s(t)\, s_m^*(t)\, dt \qquad n(t,\tau) = \int_{T_P} w(t,\tau)\, \underset{1\times1}{s_m^*(t)}\, dt$$

$$\underset{MN\times1}{y} = \sum_{k=1}^{K} \bar{a}_t(\theta_k) \otimes a_r(\theta_k) \cdot r_k + n$$

$$= A_{MN\times K}\, r_{K\times1} + n$$

若 $s(t)$ 互相正交，则 $a(\theta) = a_t(\theta) \otimes a_r(\theta)$

$$= e^{j\pi(p_m + q_n)\cos\theta_k}$$

$S_1 = 0, 2, 5, 9 \quad \times \frac{\lambda_1}{2} \quad 0, 2, 5, 9$ $\qquad M = 4$

$S_2 = 15, 25, 26, 38 \quad \times \frac{\lambda_2}{2} \quad 0, 10, 11, 23$ $\qquad K = 3$

$S_3 = 44, 50, 58, 74 \quad \times \frac{\lambda_3}{2} \quad 0, 6, 14, 30$ $\qquad P = \max\{9, 23, 30\} + 1 = 31$

B_2
31×31

$\hat{R}_{k_{M \times M}} \qquad k = 1, 2, 3$

$$\hat{R}_{P \times P} = \left(\sum_{k=1}^{K} \hat{R}_k \circ B_k \right) \circ D$$

$f_1 \; \cancel{\uparrow} \; 3$

$\overset{0}{d_1} \quad \overset{1}{d_2} \quad d_3$

$S = \{ d_i - d_j \} \longrightarrow y_{9 \times 1} = A_{M^2 \times N} x$

$0, 1, 2, 3$

$\downarrow$

$\tilde{y}_{4 \times 1}$

$f_2 \; \cancel{\uparrow} \; 6$

$\overset{0}{d_1} \quad \overset{2}{d_2} \quad d_3$

$y_{9 \times 1}$

$\downarrow$

$S = 0, 2, 4, 6 \qquad \tilde{y}_{4 \times 1}$

$$\tilde{y} = \begin{pmatrix} \tilde{y}_1 \\ \tilde{y}_2 \end{pmatrix}_{0, 1, 2, 3, 4, 6} \longrightarrow \hat{R}$$